Morimoto Lab

Bio-Inspired Backbone Geometry for Multi-Point 3D Force Estimation in Continuum Robots

Noah Jones, T.K. Morimoto

UC San Diego
JACOBS SCHOOL OF ENGINEERING

Problem

- **Motivation:** Knowledge of robot-tissue interaction forces during surgery can reduce patient injury and improve quality of surgeon's force feedback.
- **Goal:** Estimate the magnitudes and locations of multiple point forces.
- **Body-Mounted Sensing (FBGs, EM, Resistive flex):**
  - **Pros:** Very accurate **Cons:** Hard to sterilize, bulky, hard to integrate [1].
- **Visual-Based Sensing (Endoscopic cameras, fiducial tracking) :**
  - **Pros:** Can resolve multiple point forces with high accuracy [2][3], easy to sterilize.
  - **Cons:** Requires direct line-of-sight of robot body, camera space constraints.
- **Proximal-Mounted Sensing (force/torque sensors, Tendon tension sensors):**
  - **Pros:** Works in blind environments, easy to sterilize, easy to integrate [4]
  - **Cons:** Current approaches cannot resolve magnitudes and locations of multiple point forces (ill-posed inverse).

Approach

- **Idea:** We propose estimating multiple point forces using proximal sensors and variations in robot backbone geometry.
- **Inspiration:** Sensory appendages in animals (whiskers, antennae) use non-circular cross-sections and mass asymmetry to mechanically amplify weak signals—like torsion—that would otherwise be masked by dominant bending modes [5]. These geometries help overcome the ill-posed nature of multi-force sensing in straight cylindrical designs [6][7].
- **Method:** We develop an optimization framework to find backbone geometries that maximize multi-contact force estimation accuracy. The backbone is modeled as a series of elliptical cross-sections, with the major and minor radii of each segment as design variables. The objective is to find geometries with well-conditioned Jacobians—ensuring sensor noise is not amplified into large force estimation errors.

Results

- **Design Evaluation:** We compare designs numerically and graphically using manipulability ellipsoids. Superior geometries maximize the shortest axis (minimum singular value), ensuring sensitivity in all directions.

- **Experimental Validation:** For inter-force separations greater than 8 cm, the optimized design achieves median absolute errors of approximately 0.05 N, 1.0 mm, and 2.2°—corresponding to performance improvements of 74%, 67%, and 66% respectively over the cylindrical baseline. Performance gains diminish at smaller separations but remain measurable.

Significance

- **Sensing Through Shape:** Just as natural selection has shaped animals' sensory appendages to optimize perception for their specific environments, we extend this principle to design tendon-driven surgical robots with geometries tailored to the complex loading conditions they encounter.

- **Optimization:** We show how optimization can be used to find an optimal backbone design, capable of resolving multiple point-loads that conventional cylindrical backbones cannot due to inherent sensing degeneracies.

- **Hardware Testing:** We develop a custom testing rig featuring multiple PID controlled tensioning winches and a two-arm gimbal, capable of applying 3 tendon tensions, and 2 point-forces in 3D.

References

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